

COMP101  
Introduction to Programming  
2025-26  
Lecture-01  
Introduction

# COMP101 – Tutor

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# COMP101 – Components

## 1. Python programming

Procedural paradigm – not object oriented (O-O)

Code an algorithm (a problem)

## 2. Computational Thinking

‘Problem Solving’ - produce algorithms that can be coded

## 3. Analysis and design

House Style, Pseudocode, flow charts etc

## 4. Document

In-line commentary, testing

# COMP101 – Programming content

- Control statements
  - sequence – selection – iteration
- Data types, casting, variables and constants
- I/O control, numbers, string handling
- Functions (modularisation)
- Exception handling
- Data Structures
  - ‘Collections’ (1-d and 2-d arrays)

# COMP101 – not covered in assessment

- Standard algorithms
  - sorting, searching, regular expressions, path finding, set functions, hashing, recursion
- Objects and classes
- Files and streams
- Data structures
  - linked lists, stacks, queues, trees

# COMP101 – Resources

- Material:
  - Everything on Canvas
- Attendance:
  - Lectures = 3 each week      Week 1 to 10 (theory and techniques)
  - Labs            = 1 each week      Week 3 to 10 (exercises and assignments)
  - Tutorials = 1 each week      Week 3 to 8 (program design)
- Weeks 11 and 12 for support and assignment completion

# COMP101 Calendar

| Week Number | Lectures       | Labs<br>Supervised week-03 to week-09 | Tutorials<br>Supervised week-03 to week-08 | Assignments<br>7 Coursework Assessments<br>No examination |
|-------------|----------------|---------------------------------------|--------------------------------------------|-----------------------------------------------------------|
| 01          | Three Lectures | One Lab - unsupervised                | One Tutorial – unsupervised                | No assessment                                             |
| 02          | Three Lectures | One Lab – unsupervised                | One Tutorial - unsupervised                | Ass-01 release (12%)                                      |
| 03          | Three Lectures | One Lab – supervised                  | One Tutorial – supervised                  | Ass-01 submit<br>Ass-02 release (12%)                     |
| 04          | Three Lectures | One Lab – supervised                  | One Tutorial - supervised                  | Ass-02 submit<br>Ass-03 release (13%)                     |
| 05          | Three Lectures | One Lab – supervised                  | One Tutorial - supervised                  | Ass-03 submit                                             |
| 06          | Three Lectures | One Lab – supervised                  | One Tutorial – supervised                  | Ass-04 release (13%)                                      |
| 07          | Three Lectures | One Lab – supervised                  | One Tutorial - supervised                  | Ass-04 submit<br>Ass-05 release (16%)                     |
| 08          | Three Lectures | One Lab – supervised                  | One Tutorial – supervised                  | Ass-06 release (16%)                                      |
| 09          | Three Lectures | One Lab – supervised                  | No tutorial                                | Ass-05 submit                                             |
| 10          | Three Lectures | No lab                                | No tutorial                                | Ass-06 submit<br>Ass-07 release (18%)                     |
| 11          | No Lectures    | No lab                                | No tutorial                                |                                                           |
| 12          | No Lectures    | No lab                                | No tutorial                                | Ass-07 submit                                             |

# COMP101 or COMP105? Decide by week-02

- COMP101

- No or limited programming experience
- Uses procedural programming
  - no object-orientation or any other paradigm
- Python
  - High-level, interpreted language, very good (and easier) for learning

- COMP105

- Can already program to some degree of confidence
  - No benefit being in COMP101
- Let us know in week-02 if you need to swap to COMP105
  - John Fearnley [John.Fearnley@liverpool.ac.uk](mailto:John.Fearnley@liverpool.ac.uk)
  - Keith Dures [dures@liverpool.ac.uk](mailto:dures@liverpool.ac.uk)



# COMP101

## Assumed knowledge of programming

# COMP101 aims

- Develop into a programmer (not a ‘coder’) based on robust coding
  - Employ rigour and discipline in programming, computational thinking, analysis and design and documentation
  - Accurate and critically evaluated testing
  - Structured design methodologies
  - Critically applied computational thinking
- Python used to learn to program in the ‘procedural paradigm’
  - COMP101 is ‘Introduction to Programming’
  - Python is the tool – the module isn’t ‘Programming in Python’